MIL-E-63124A(AR)
17 December 1979
SUPERSEDING
MIL-E-63124(AR)
7 September 1977

MILITARY SPECIFICATION

ELECTRONICS MODULE ASSEMBLY

This Specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Specification covers the performance and acceptance inspection requirements for the Electronics Module Assembly associated with the M130 Aircraft General Purpose Dispenser.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC: 1095

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-781 - Reliability Tests: Exponential Distribution
MIL-STD-810 - Environmental Test Methods
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures for

DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9311431 - Electronics Module Assembly
9321259 - Packing and Marking for Box, Exterior for Electronics Module Assembly
9313708 - Packing and Marking for Drum, Metal, Shipping for M130 Aircraft General Purpose Dispenser

U.S. ARMY MATERIEL DEVELOPMENT AND READINESS COMMAND

19-48-4138/109-20 PA1000 - Unitization Procedures for Boxed Ammunition and Components on Skidded Bases

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9311431 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Functioning. The electronics module, having successfully completed burn-in (See 4.5.7), shall perform as specified herein.

3.3.1 Input. The electronics module shall respond to an external 22.5 to 30.0 volt direct current pulse on any of four inputs.

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3.3.2 Arm. The electronics module shall respond to input signals only when a22.5 to 30.0 volt direct current (d.c.) ARM signal is present.

3.3.3 Dispense output. The electronics module shall provide two outputs of 20.0 to 30.0 volt d.c. current, a chaff dispense signal and a flare dispense signal. These outputs shall be initiated by the respective input signals.

3.3.3.1 Signal dispense output. In the case of chaff dispense input, the electronics module shall provide only one dispense output signal.

3.3.3.2 Chaff program. In response to the chaff program input, the electronics module shall provide chaff dispense output signals in accordance with the setting or the position of the chaff program select switches (see Table A).

3.3.3.3 Ripple Fire. In response to the flare ripple fire command, the electronics module shall generate flare dispense output signals in sequential order sufficient to dispense all remaining flares within four seconds max.

3.3.4 Flare misfire detection logic. If the electronics module does not receive a flare detection signal from the dispenser assembly, it shall be limited to three flare dispense actions only for each flare dispense input signal. If the electronics module does receive a flare detection signal, it shall not generate additional dispense signals.

3.3.5 Chaff/flare count signal. A chaff count signal or a flare count signal shall be generated each time a chaff or flare is dispensed. These signals shall be 20.0 to 30.0 volts d.c.

3.4 Environmental requirements. The electronics module shall perform as specified in 3.3 after exposure to the environments specified herein.

3.4.1 Aircraft vibration. The electronics module shall perform all functions satisfactorily after being subjected to the vibration requirements of MIL-STD-810, Test Method 514.2, Procedure IIc. The electronics module shall be vibrated along three mutually perpendicular axes as shown in Figure 1. The payload module shall be loaded with 30 inert flares for this test.

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TABLE A. CHAFF EJECTION MODES

SELECTOR SWITCH	FUNCTION	SETTING AVAILABLE TOLERANCE = ± 20%
BURST COUNT	Sets number of payloads fired per salvo.	1, 2, 3, 4, 6, or 8
BURST INTERVAL	Sets time interval in seconds between payload firing for Salvos in which more than one payload is fired.	0.1, 0.2, 0.3, or 0.4 seconds
SALVO COUNT	Sets number of salvos fired per program. The "C" setting causes a continuous number of salvos to be fired until the chaff payload is emptied.	1, 2, 4, 8, or C
SALVO INTERVAL	Sets time interval in seconds between the last payload fired in the previous salvo and the first payload fired in the following salvo. The "R" setting causes the salvo interval to vary from salvo to salvo in a pseudo random manner based upon a factory set sequence.	1, 2, 3, 4, 5, 8, or R

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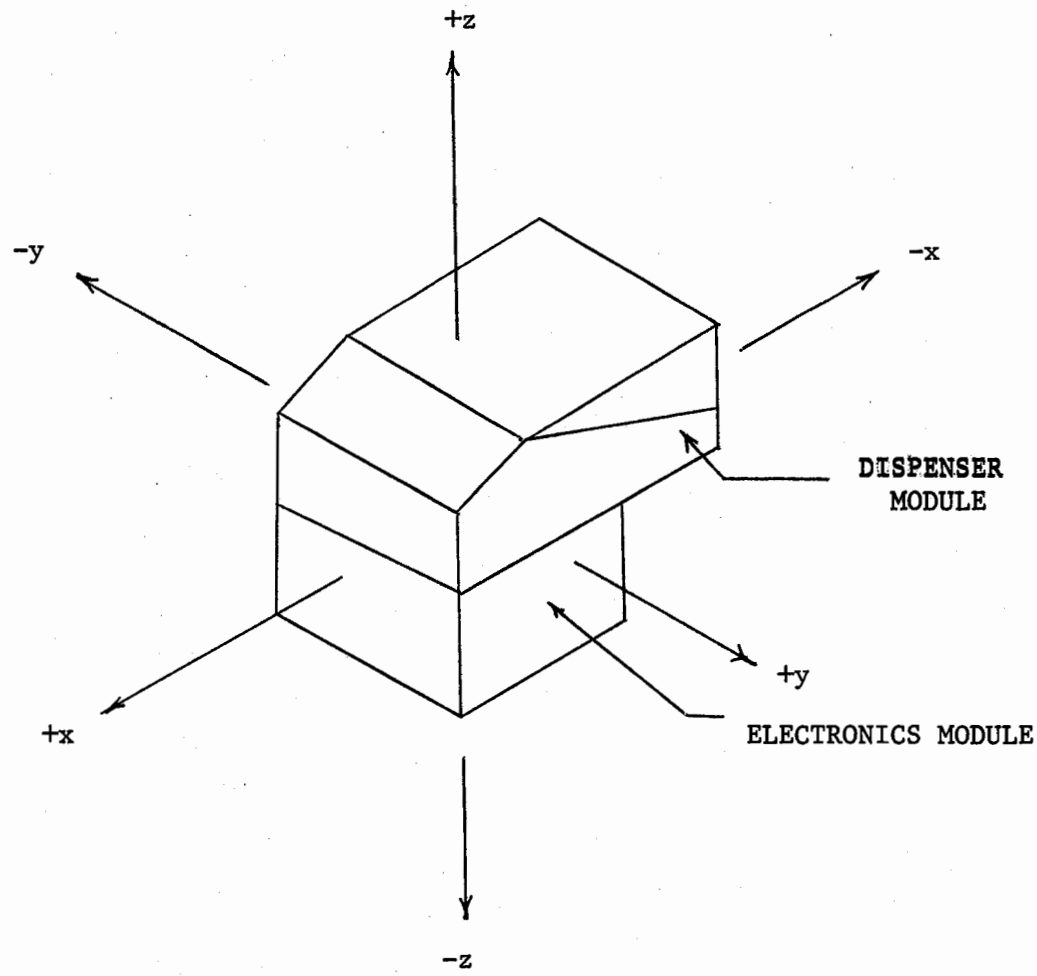


Figure 1. Dispenser System Test Axis Orientation

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3.4.2 Transportation vibration. The electronics module shall perform all functions satisfactorily after being subjected to the transportation vibration requirements of MIL-STD-810, Test Method 514.2, Procedure X using curve AW of Figure 514.2-7 (change curve to 0.8" displacement (double amplitude) between 5&6 Hz). The electronics module shall be vibrated along three mutually perpendicular axes as shown in Figure 1. The payload module shall be loaded with 30 inert flares for this test.

3.4.3 Humidity. The electroncis module shall perform all functions satisfactorily after being subjected to MIL-STD-810, Test Method 507.1, Procedure I. (10 days)

3.4.4 High temperature. The electronics module shall perform all functions satisfactorily at +54°C according to MIL-STD-810, Test Method 501.1.

3.4.5 Low temperature. The electronics module shall perform all functions satisfactorily at -54°C according to MIL-STD-810, Test Method 502.1.

3.5. Wiring and soldering. Soldering of electrical wires, cables, connectors, components and assemblies shall be performed in a thorough, workmanlike manner in accordance with MIL-STD-1460. (Also See 3.7)

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Cleanliness. Work areas and tools shall be maintained in a clean and orderly condition. All dirt, grease, oil, solder splatter, chips and other contaminating foreign matter shall be promptly removed. Thermal shunts and anti-wicking tools should be cleaned frequently. Cleaned component leads, terminals, etc., shall not be handled with bare hands. The use of white lint free gloves is recommended. Smoking, eating or drinking shall not be permitted at any soldering bench or station.

3.8 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing, and assembling operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly, nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Electronics Module	9311431	2

4.3.2 Inspections to be performed. See MIL-A-48078 and paragraphs 4.4.2.1 through 4.4.2.18 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-E-63124A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Electronics Module Assembly				9311431
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for Defects	2		3.2	4.4.2.14(a)
	Burn-in	2		3.3	4.5.7
	Functioning Test	2		3.3	4.5.1
	Transportation Vibration Test	2		3.4.2	4.5.5
	Aircraft Vibration Test	2		3.4.1	4.5.4
	Humidity Test	2		3.4.3	4.5.6
	Performance at High Temperature Test	2		3.4.4	4.5.2
	Performance at Low Temperature Test	2		3.4.5	4.5.3
	Examination for Defects	2		3.2	4.4.2.14(a)
	(NOTE: The above tests shall be performed in the order indicated. The two units submitted to environmental testing shall not be returned to the deliverable lot.)				

NOTES:

(a) Omit Major 101 from 4.4.2.14 for First Article Inspection.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. (See MIL-A-48078). Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311565
4.4.2.1	Housing Assembly, Electronics				NEXT HIGHER ASSEMBLY 9311477
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of mounting holes improper		0.40%	3.2	Gage
102	Brazing incomplete or inadequate		0.40%	3.2	Visual
103	Any part missing, damaged or improperly assembled		0.40%	3.2	Visual
Minor					
201	Protective finish missing or scratched		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311527
4.4.2.2	Printed Wiring Board Assembly, IC, Salvo, Chaff				NEXT HIGHER ASSEMBLY 9311473
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or inadequate		0.40%	3.2	Visual
103	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Printed Wiring Board Assembly, Discrete, Salvo, Chaff				9311524
					NEXT HIGHER ASSEMBLY
					9311473
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or improper		0.40%	3.2	Visual
103	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Printed Wiring Board Assembly, Discrete, Burst, C/F		9311521
			NEXT HIGHER ASSEMBLY
			9311473

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or improper		0.40%	3.2	Visual
103	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Printed Wiring Board Assembly, IC, Burst, Chaff				9311518
					NEXT HIGHER ASSEMBLY
					9311473
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or improper		0.40%	3.2	Visual
103	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311477
4.4.2.6	Housing Assembly, Electronics Module				NEXT HIGHER ASSEMBLY 9311431
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Paint missing or improper		0.40%	3.2	Visual
102	Marking missing or improper		0.40%	3.2	Visual
103	Rubber adhesive missing, inadequate or improperly applied		0.40%	3.2	Visual
Minor					
201	Protective finish missing or scratched		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311473
4.4.2.7	Module Assembly, Programmer, Chaff				NEXT HIGHER ASSEMBLY 9311429
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Alignment of knobs improper		0.40%	3.2	Visual
103	Sealant missing from threaded surfaces		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Printed Wiring Board Assembly, Front Plane				9311467
					NEXT HIGHER ASSEMBLY
					9311473
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or improper		0.40%	3.2	Visual
103	Switch stop pins missing or improperly installed		0.40%	3.2	Visual
104	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Printed Wiring Board Assembly, Interconnect				9311461
					NEXT HIGHER ASSEMBLY
					9311429
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or improper		0.40%	3.2	Visual
103	Pins not trimmed in accordance with drawing		0.40%	3.2	Visual
104	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311458
4.4.2.10	Printed Wiring Board Assembly, Flare Detector				NEXT HIGHER ASSEMBLY 9311429
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or improper		0.40%	3.2	Visual
103	Polarizing plug missing or improperly located		0.40%	3.2	Visual
104	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311455
4.4.2.11	Printed Wiring Board Assembly, Power Supply Number 2				NEXT HIGHER ASSEMBLY 9311429
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or improper		0.40%	3.2	Visual
103	Polarizing plug missing or improperly located		0.40%	3.2	Visual
104	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Printed Wiring Board Assembly, Power Supply #1				9311452
					NEXT HIGHER ASSEMBLY
					9311429
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Insulating compound missing or improper		0.40%	3.2	Visual
103	Sealing compound missing from threaded surface		0.40%	3.2	Visual
104	Soldering improper or inadequate		100%	3.5	4.4.3.7/Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311429
4.4.2.13	Programmer, Module				NEXT HIGHER ASSEMBLY 9311431
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.2	Visual
102	Sealing compound missing from threaded surfaces		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Module, Electronics		9311431
			NEXT HIGHER ASSEMBLY 9311430

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Burn-in		100%	3.3	4.5.7
102	Functioning Test		100%	3.3	4.5.1
103	Any part missing, damaged or improperly assembled		0.40%	3.2	Visual
104	Identification plate missing, improperly annotated or improperly attached		0.40%	3.2	Visual
105	Sealing compound missing from threaded surfaces.		0.40%	3.2	Visual
Minor					
201	Protective finish scratched		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Packing and Marking for Drum, Metal Shipping for M130 Aircraft General Purpose Dispenser				9313708
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cushions missing or not secured		0.40%	3.2	Visual
102	Contents not thoroughly dry		0.40%	3.2	Visual
103	Any component missing or damaged		0.40%	3.2	Visual
Minor					
201	Markings missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Fillers insufficient, loose contents		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Sealed Barrier Bag, Prior to Packing in Box	1 OF 1	9321959
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Waterproofness of barrier bag destroyed by puncture or separation of heat seal		0.40%	3.2	Visaul
102	All excess air not exhuasted from bag prior to sealing		0.40%	3.2	Visual
103	Evidence of crushed box NOTE: Packing should be opened if there is evidence of crushing.		0.40%	3.2	Visual
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.2	Visual
202	Marking of bag misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Packing and Marking for Box, Fiberboard for Electronics Module Assembly				9321260
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any packing component missing		0.40%	3.2	Visual
102	Markings missing, incomplete or illegible		0.40%	3.2	Visual
Minor					
201	Box not securely closed		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63124A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Packing and Marking for Box, Exterior for Electronics Module Assembly		9321259
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any barrier bag or filler missing		0.40%	3.2	Visual
102	Exterior markings missing, incomplete or illegible		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual
202	Box not securely closed		0.65%	3.2	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing

4.4.3.1 Functioning. Each electronics module, after compliance with 4.5.7, shall be subjected to this test. The test shall be performed as specified in 4.5.1 using equipment in accordance with 4.4.4. If any unit fails to comply with the requirements of 3.3, it shall be classified defective and removed from the lot.

4.4.3.2 High temperature. Each electronics module submitted for first article testing shall be subjected to functioning at high temperature. The test shall be performed after Humidity testing (see 4.4.3.6) as specified in 4.5.2 using equipment in accordance with 4.4.4. If any unit fails to comply with the requirements of 3.4.4, the first article sample shall be rejected.

4.4.3.3 Low temperature. Each electronics module submitted for first article shall be subjected to functioning at low temperature. The test shall be performed as specified in 4.5.3 using equipment in accordance with 4.4.4. If any unit fails to comply with the requirements of 3.4.5, the first article sample shall be rejected.

4.4.3.4 Aircraft vibration. Each electronics module submitted for first article testing shall be subjected to this test. The test shall be performed as specified in 4.5.4 using equipment in accordance with 4.4.4. If any unit fails to comply with the requirements of 3.4.1, the first article sample shall be rejected.

4.4.3.5 Transportation vibration. Each electronics module submitted for first article testing shall be subjected to this test. The test shall be performed as specified in 4.5.5 using equipment in accordance with 4.4.4. If any unit fails to comply with the requirements of 3.4.2, the first article sample shall be rejected.

4.4.3.6 Humidity. Each electronics module submitted for first article testing shall be subjected to this test. The test shall be performed as specified in 4.5.6 using equipment in accordance with 4.4.4. If any unit fails to comply with the requirements of 3.4.3 the first article sample shall be rejected. Following this test, the electronics assembly shall be subjected to the functioning at high and low temperature (See 4.4.3.2).

4.4.3.7 Soldering. Each electronics module shall be inspected for soldering in accordance with 3.5. If any wiring or printed circuit card assembly fails to comply with the requirements of 3.5, the unit shall be classified defective and removed from the lot.

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4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column of the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.3 herein.

4.5 Test methods and procedures

4.5.1 Functioning. The electronics module, excluding those conditioned at high or low temperature, shall be conditioned for four (4) hours minimum at 25 ± 5°C (ambient). The unit shall be installed in a test setup that will provide the minimum input direct current as specified in 3.3.1 into each of the four input terminals, and measure the voltage at the respective output terminal. Dispense signals shall be initiated in each of the payload select modes to verify conformance to 3.3.1, 3.3.2, 3.3.3, and 3.3.5. A flare dispense signal shall be provided both with and without 6.5 VDC maximum flare detect signal to verify conformance with 3.3.4. Repeat this test with the maximum input direct current as specified in 3.3.1 applied to the electronics module.

4.5.2 High temperature. After conditioning at 54° ± 2°C according to MIL-STD-810, Test Method 501.1, the electronics module shall be subjected to the functioning test of 4.5.1 at high temperature.

4.5.3 Low temperature. - After conditioning at -54° ± 2°C according to MIL-STD-810, Test Method 502.1, the electronics module shall be subjected to the functioning test of 4.5.1 at low temperature.

4.5.4 Aircraft vibration. This test shall be performed as specified in MIL-STD-810, Test Method 514.2, Procedure IIc assuming the number of missions to be greater than 100. Following this test, examine each electronics module for evidence of damage and submit each item to the functioning test of 4.5.1.

4.5.5 Transportation vibration. This test shall be performed as specified in MIL-STD-810, Method 514.2, Procedure X using curve AW of Figure 514.2-7 (change curve to 0.8" displacement (double amplitude) between 5&6 Hz). Following this test, examine each electronics module for evidence of damage and submit each item to the functioning test of 4.5.1.

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4.5.6 Humidity. This test shall be performed as specified in MIL-STD-810, Test Method 507.1, Procedure I. (10 days)

4.5.7 Burn-in. Each electronics module shall successfully complete a minimum of two cycles of burn-in in accordance with MIL-STD-781. The temperature limits shall be 71 ± 2°C (high), 25 ± 2°C (ambient) and -54 ± 2°C (low) with a vibration peak acceleration value of 2.2g ± 10% at a frequency between 20 and 60 hz and a nominal voltage of 28 ± 2 VDC. Actual test duration shall be determined by thermal survey in accordance wth MIL-STD-781. If desired, only the Module Programmer (Dwg. 9311429) of the test item need be subjected to burn-in.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Preservation and packaging shall be as specified on Drawing 9313708 when shipped as part of the M130 Aircraft General Purpose Dispenser, or Drawing 9321259 when shipped as a repair part.

5.2 Packing.

5.2.1 Level A. Packing shall be as specified on Drawing 9313708 when shipped as part of the M130 Aircraft General Purpose Dispenser, or Drawing 9321259 when shipped as a repair part.

5.3 Marking.

5.3.1 Level A. Marking shall be as specified on Drawing 9313708 when shipped as part of the M130 Dispenser, or Drawing 9321259 when shipped as a repair part.

5.4 Unitization.

5.4.1 Level A. Only when shipped as part of the M130 Dispenser System, the electronics module assembly in the drum shall be unitized in accordance with Dwg. 19-48-4138/109-20 PA1000 (DARCOM Ammo Center).

6. NOTES

6.1 Intended use. This electronics module is part of the Aircraft General Purpose Dispenser, M130.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078) Submit equipment designs as required to Commander, U.S Army Armament Research and Development Command, DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, U.S Army Armament Research and Development Command, Attn: DRDAR-QAA-M, Dover, New Jersey 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARRADCOM.

6.6 Changes from previous issue. Asterisks are not used in this revision with respect to the previous issue, due to the extensiveness of the changes.

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Project No: 1095-A127

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